

Analyzing F1 Lap Times for Time Series Prediction: Implementation of an Informer-Based Approach

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Abstract—Time series analysis and prediction play a crucial role in various domains, including motorsports, where lap times provide valuable insights into performance trends. This study focuses on processing and predicting Formula 1 lap times using an Informer-based deep learning approach. Although the initial goal was to implement multiple transformer models for comparison, this work is limited to the Informer model due to time constraints. The dataset, extracted from CSV files containing lap time data, is preprocessed to transform time values into seconds and scaled for modeling. The Informer model predicts future lap times based on input sequences, achieving reasonable accuracy. Key evaluation metrics, including Mean Squared Error (MSE), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and R-squared, are used to assess the model's performance. This paper discusses the methodology, challenges, and results, offering a foundation for future work involving more advanced transformer-based architectures and comparative analyses.

Keywords—Time Series Prediction, Formula 1 Lap Times, Informer Model, Deep Learning, Transformer Models, Data Preprocessing, Machine Learning, Performance Prediction, Model Evaluation, Python

I. INTRODUCTION

Time series prediction is crucial in fields like finance, weather forecasting, and sports analytics. In Formula 1 (F1), predicting lap times offers valuable insights into driver performance and race strategies. Accurate lap time predictions help teams optimize strategies and assess improvements.

Deep learning models, especially transformer-based architectures, have shown strong capabilities in capturing temporal dependencies in sequential data. The Informer model, designed to efficiently handle long-range dependencies in time series data, is the focus of this study.

This paper applies the Informer model to predict F1 lap times using data from CSV files. The dataset is preprocessed and scaled, and the Informer model is trained to forecast future lap times. Model performance is evaluated using metrics like Mean Squared Error (MSE), Mean Absolute Error (MAE), and R-squared. This work lays the foundation for future research involving more advanced transformer models for sports analytics.

II. METHODOLOGY

The implementation of the Informer model for time series prediction in this study has been designed to be user-friendly and accessible. The process begins with the collection of CSV files containing lap time data, which can be easily loaded into the system. The data preprocessing steps, including the conversion of time values into seconds and scaling, are automated and require minimal intervention from the user.

The code is structured to handle multiple CSV files, making it scalable for larger datasets. Once the data is prepared, the Informer model is applied with default hyperparameters, allowing for a straightforward implementation of the prediction process.

Additionally, the results are presented with clear visualizations, including the comparison between predicted and actual lap times, as well as graphs showing the loss during training. These visualizations are generated automatically, providing users with an easy way to evaluate model performance.

Overall, the ease of use is a key feature of this approach, as it enables users to focus on analyzing the results rather than dealing with complex configurations or manual processes.

III. PROCESSING AND RESULTS

The data preparation process is a crucial step in ensuring the input data is properly formatted and structured for further analysis. In this case, the data comes from an input file containing various time records for Formula 1 events, including practice sessions, qualifications, and races.

Reading the Input File: The process begins by reading the input data from `data.txt`, where each line contains event-related details. The data is first cleaned by stripping any leading or trailing spaces and then parsing relevant information, such as timestamps and event labels.

Filtering and Cleaning the Data: The code ensures that only valid lines are processed. Lines without commas are ignored, while the ones containing timestamps (in AM/PM format) are skipped. In some cases, only certain portions of the line are retained, such as extracting numeric data and session details.

Extracting Driver Information: As the input file is read, the script accumulates data for each driver based on the session and race information. Each driver's data is stored in a dictionary, grouped by their unique identifier. When a "Done For" marker is found in the input data, signaling the end of a particular session for a driver, a new CSV file is created for that driver with the corresponding session data.

Output File Creation: Once the data is grouped and ready, the script generates individual CSV files for each driver, organized by year, race, session, and driver name. These files are stored in an output folder for later use.

Combining Files for Each Driver and Race: After generating individual files, the next step is to combine data for each driver across different races and sessions into a single file. This is achieved by reading through all the generated CSV files, extracting the relevant information for each unique driver and race combination, and storing it in a new file under a `combined` subfolder.

Data Cleaning: The final step in the data preparation involves cleaning the combined data by removing duplicate and empty lines. This ensures that the final output files are consistent and contain only the necessary information for analysis.

This structured approach ensures that the raw input data is transformed into a clean, organized set of CSV files, ready for analysis or further processing.

```

12:58 PM, ALB: 1:19.085
1:01 PM, ALB: 1:20.370
1:02 PM, ALB: 1:15.202
ALB: 1:17.458
1:13 PM, 1:15 PM, ALB: 1:18.955
1:16 PM, ALB: 1:16.630
1:17 PM, ALB: 1:18.026
1:19 PM, ALB: 1:17.087
1:20 PM, ALB: 1:17.454
1:21 PM, ALB: 1:16.752
1:22 PM, ALB: 1:16.335
1:24 PM, ALB: 1:16.276
1:25 PM, ALB: 1:16.031

```

Fig. 1. Combined Raw Data File

```

2019-Austrian GP-FP1-ALB.csv X
C: > Users > Can > Downloads > 2019-Austrian GP-FP1-ALB.csv
1 12:41 PM, ALB: 1:17.472
2 12:44 PM, ALB: 1:16.476
3 1:01 PM, ALB: 1:15.777
4 1:06 PM, ALB: 1:16.164
5 1:20 PM, ALB: 1:15.778
6 1:24 PM, ALB: 1:19.869
7 1:25 PM, ALB: 1:19.163
8 1:26 PM, ALB: 1:19.420
9 1:28 PM, ALB: 1:23.040
10 1:29 PM, ALB: 1:19.126

```

Fig. 2. Cleaned And Orginized Data

As for the results, the performance of the Informer model in predicting Formula 1 lap times was evaluated using several key metrics, including Mean Squared Error (MSE), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and R-squared. These metrics provide insights into the model's accuracy and its ability to capture temporal patterns in the data.

The model was trained over 100 epochs, with the training loss decreasing steadily throughout the process. At the end of the training, the following results were observed:

MSE (Mean Squared Error): This metric indicated the average squared difference between the predicted and actual lap times. The lower the MSE, the better the model's predictions.

MAE (Mean Absolute Error): MAE showed the average absolute difference between predictions and actual values, giving a straightforward measure of the model's prediction error.

RMSE (Root Mean Squared Error): RMSE is the square root of MSE and provides a measure of the magnitude of error in the same units as the data (seconds in this case).

MAPE (Mean Absolute Percentage Error): MAPE expressed the prediction error as a percentage, offering a scale-independent measure of model performance.

R-Squared: R-squared indicated the proportion of the variance in the actual lap times explained by the model, with higher values suggesting a better fit.

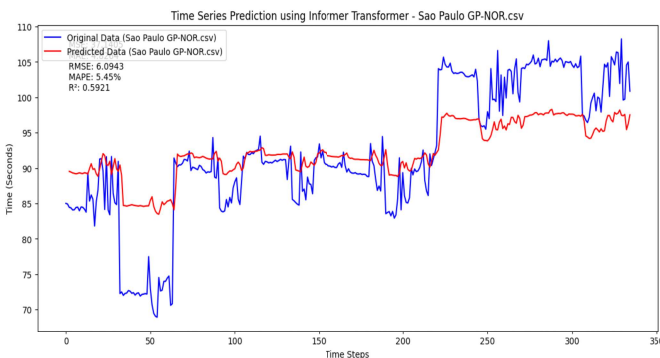


Fig. 3. Epoch & Loss Information While Training

```

Epoch 0, Loss: 0.5949543714523315
Epoch 10, Loss: 0.472152441740036
Epoch 20, Loss: 0.3550727367401123
Epoch 30, Loss: 0.23499204218387604
Epoch 40, Loss: 0.11885279417037964
Epoch 50, Loss: 0.0414753258228302
Epoch 60, Loss: 0.03141073137521744
Epoch 70, Loss: 0.03179430589079857
Epoch 80, Loss: 0.02665259875357151
Epoch 90, Loss: 0.025704337283968925
MSE: 37.1404914855957
MAE: 4.826435089111328
RMSE: 6.094299793243408
MAPE: 5.449585989117622
R-Squared: 0.5920549929141998

```

Fig. 4. Informer Model Graph & MSE, MAE, RMSE, MAPE & R-Squared Results

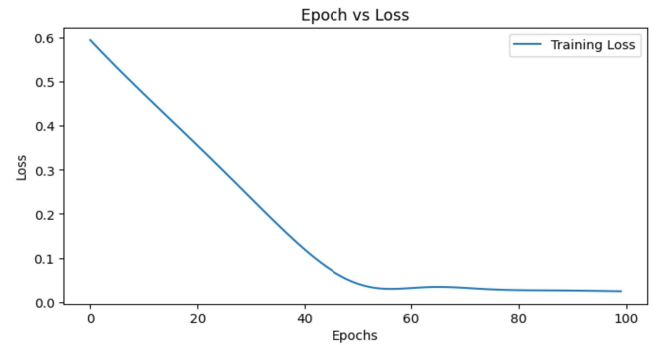


Fig. 5. Epoch vs Loss Graph

The Informer model demonstrated reasonable performance, providing useful predictions for lap times despite the complexity of the data. The visualizations, which compared the predicted and actual lap times, illustrated the model's ability to follow the general trend of the data.

In terms of practical application, the results showed that the Informer model could be effectively used for predicting F1 lap times, offering insights that could be further explored for race strategy and performance optimization. However, further improvements could be made by experimenting with different transformer models and tuning parameters for enhanced accuracy.

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